

# Riley Brown

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## Education

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### University of Nebraska-Lincoln

Aug 2023–May 2027

*B.S. Mathematics and Data Science, Minor in Computer Science*

(expected)  
Lincoln, NE

- GPA 3.92/4.00, University Honors Program
- President of Pi Mu Epsilon and Math Club; Putnam Exam Practice

### Budapest Semesters in Mathematics

Jan 2026–Jun 2026

*Study abroad, mathematics*

Budapest, Hungary

## Research Experience

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### Undergraduate Researcher - Quantum LDPC Codes

Fall 2026

*University of Nebraska-Lincoln* advisor: *Professor Christine Kelley*

Lincoln, NE

- Testing whether collision-free geodesic-flow operators on Ramanujan complexes can sharpen iterative-decoder thresholds for quantum LDPC codes

### Research Assistant — pseudorandomness

Summer 2026

*North Carolina State University* advisor: *Professor Chin Ho Lee*

Raleigh, NC

- Studied Ta-Shma's epsilon-balanced code construction with Prof. Chin Ho Lee as a research partner
- Found simpler constructions and minor improvements to the Hsieh-Mohanty-Zheng construction

### Research Assistant, t-uniform hypergraph degree sequences and FPT algorithms

Spring 2026

*Budapest Semesters in Mathematics* advisor: *Professor Istvan Miklos*

Budapest, Hungary

- Proved t-uniform hypergraphicality is FPT in  $(k, t)$  for compressed degree sequences, extending known bounded-range results
- Reduced realizability to integer programming feasibility via bounded-dimension spectra and balancing hinge-flips

### Summer Researcher — algorithm design

Summer 2025

*Harvey Mudd College* advisor: *Professor Tim Randolph*

Claremont, CA

- Improved exponential-time algorithms for sumset recognition, an NP-complete problem in additive combinatorics (team of four)
- Sharpened branching strategies and base-case parameters for graph-theoretic reductions

### Investigating the Flow of Opinions in a Diagonal Grid Graph Using a Modified Deffuant-Weisbuch Model

Fall 2024

*University of Nebraska-Lincoln, Directed Reading Program* advisor: *Nick White*

Lincoln, NE

- Extended the project beyond its initial reading goals by implementing a modified Deffuant-Weisbuch model in Python

- Ran systematic parameter variation experiments showing that opinions invariably converge to consensus on diagonal grid graphs
- Proved the discovered properties and identified future directions on alternative graph topologies

### **Undergraduate Researcher — tiling theory**

Summer 2024

*University of Nebraska-Lincoln* advisor: *Professor Tri Lai*

Lincoln, NE

- Applied Ciucu’s Factorization Theorem to decompose tiling enumeration problems into tractable subproblems, working up through independent study of survey papers
- Used Randall paths to determine hexagon-half parameters based on hole parity along the symmetric axis
- Computed exact tiling counts by applying the Lindstrom-Gessel-Viennot theorem to determinants of binomial coefficient matrices

## **Preprints and Publications**

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### **Fixed-Parameter Tractability of t-Uniform Hypergraphicality**

**Riley Brown**, Istvan Miklos

*arXiv:2606.08523*, 2026

arXiv

## **Awards and Honors**

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### **D&F Eastman Scholarship**

*Mathematics Department recognition*

### **Regents Scholarship**

### **Nebraska Career Scholarship**

### **Dean’s List**

*all semesters*

### **University Honors Program**

## **Talks**

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### **Algorithms for the Sumset Recognition Problem**

2025

*Harvey Mudd College REU Symposium*

### **Combinatorial Game Theory**

2025

*UNL Directed Reading Program Symposium*

### **Mathematics at UNL**

2025

*Marshall Recruitment Talk for High School Students*

### **The Modified Deffuant-Weisbuch Model**

2024

*UNL Directed Reading Program Symposium*

### **Lozenge Tilings of a Hexagon with Holes**

2024

*UNL Math Club*

## Teaching

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|-------------------------------------------------------------------------------------------------------------------------|------|
| <b>Teaching Assistant, Calculus I</b><br><i>University of Nebraska-Lincoln</i>                                          | 2025 |
| <b>Tap Dance Instructor</b><br><i>Chase the Spirit of Dance, Lincoln NE and Create 308, Grand Island NE (2018-2024)</i> | 2024 |
| <b>Teaching Assistant, Calculus II</b>                                                                                  |      |
| <b>Learning Assistant, Analysis</b>                                                                                     |      |
| <b>Teaching Assistant, Honors Calculus II</b>                                                                           |      |

## Activities

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| <b>President, Pi Mu Epsilon</b><br><i>University of Nebraska-Lincoln</i>                                                                | 2026 |
| <b>President, Math Club</b><br><i>University of Nebraska-Lincoln</i>                                                                    | 2026 |
| <b>Directed Reading Program (combinatorial game theory, measure-theoretic probability, opinion dynamics)</b><br><i>with Noah Walker</i> | 2024 |
| <b>Putnam Exam Practice</b><br><i>University of Nebraska-Lincoln</i>                                                                    | 2024 |
| <b>Quiz Bowl Club</b><br><i>literature, modern art, mathematics, computer science</i>                                                   | 2024 |
| <b>Poker Club</b><br><i>University of Nebraska-Lincoln</i>                                                                              | 2024 |

## Projects

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| <b>leanverify</b><br><i>Lean 4 proof checker against Mathlib: rejects sorries and smuggled axioms</i>                                                                                                                    | python, lean 4 · 2026 |
| <b>Poker Trainer</b><br><i>A terminal and Tkinter drill trainer for poker skills: pot odds, combos, equity, bet sizing, push/fold and ICM, with an adaptive scheduler that weights drills by weakness and staleness.</i> | python · 2026         |
| <b>Deep learning for patient-safety monitoring</b><br><i>Parameter tuning and data analysis for Ocuvera to classify unattended hospital bed exits</i>                                                                    | python, deep learning |

## Selected Coursework

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**Mathematics.** Bijective Partition Theory, Discrete Mathematics, Matroid Theory, Advanced Combinatorics, Conjecture and Proof, Discrete and Convex Geometry, Graph Theory, Group Theory, Probability Theory, Stochastic Processes, Elementary Analysis, Intro to Modern Algebra, Linear Algebra, Differential Equations, Multivariable Calculus, Math in the City, **Algebra I**, Coding Theory

**Computer Science. Development and Analysis of Algorithms,** Automata, Computation, and Language, Data Structures and Algorithms, Data Analysis, Math for Machine Learning, Generative AI, Computational Game Theory

## **Skills**

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**Programming.** Python, MATLAB, Java, SQL, LaTeX, Lean 4